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Antigenically stealthed oncolytic viruses

Abstract

Provided herein are antigenically stealthed HSV particles, and methods of use of the antigenically-stealthed HSV particles.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is the United States national phase of International Application No. PCT/US2019/034914 filed May 31, 2019, and claims the benefit of U.S. Provisional Patent Application No. 62/678,737 filed May 31, 2018, the disclosures of which are hereby incorporated by reference in their entirety.

(1) The Sequence Listing associated with this application is filed in electronic format via EFS-Web and is hereby incorporated by reference into the specification in its entirety. The name of the text file containing the Sequence Listing is 6527_2005635_ST25.txt. The size of the text file is 31,820 bytes, and the text file was created on Nov. 17, 2020.

(2) Oncolytic viruses (OVs) have been created from a wide variety of virus species with considerable effort devoted to achieving vector safety and efficacy. Many have been tested in human clinical trials with varying success. Vector delivery has largely relied on intratumoral inoculation without restricting off-target cell infection. A limitation of intratumoral inoculation is that many tumors cannot be inoculated, for instance due to the size or location of the tumor, and further, disseminated metastasized cells and micrometastases often cannot easily be treated locally unless individually identifiable and limited in number.

(3) We have previously developed oncolytic herpes simplex virus (oHSV) vectors that uniquely require a tumor-associated receptor for infection, such as the epidermal growth factor receptor (EGFR) or its tumor-specific variant, EGFRvIII (see, e.g., Uchida, H. et al. Effective treatment of an orthotopic xenograft model of human glioblastoma using an EGFR-retargeted oncolytic herpes simplex virus. *Mol Ther* 21, 561-569, doi:10.1038/mt.2012.211 (2013)). The engineering of retargeted oHSV has also been described in Campadelli-Fiume et al. (Retargeting strategies for oncolytic herpes simplex viruses. *Viruses* 8, 63, doi:10.3390/v8030063 (2016)). These “retargeted” oHSV can home to tumors bearing the target receptor in immune-compromised mice, but retargeted vectors have not yet been tested in patients or HSV-immune animal tumor models and it is likely that anti-HSV immunity, prevalent in the human population, will limit the effectiveness of oHSV tumor therapies.

(4) As such, a substantial impediment to use of oHSVs is host immunity to HSV and, in particular, an adaptive immune response to antigens, e.g. epitopes or antigenic determinants, present in the

oHSV particle. Sera from HSV-immune animals and patients effectively neutralize HSV by antibody-mediated mechanisms (see, e.g., Cairns, T. M., et al. 2015. Patient-specific neutralizing antibody responses to herpes simplex virus are attributed to epitopes on gD, gB, or both and can be type specific. *J Virol* 89:9213-9231). Major targets of these antibodies are two viral envelope glycoproteins, gD and gB, that are essential for HSV entry into cells. Adaptive host immunity can be developed either from an earlier HSV infection with virions comprising cross-reactive epitopes, or from earlier oHSV treatments with the same, or antigenically cross-reactive virus particles. Consequently, dissemination of virions injected systemically or replicated at the injection (tumor) site, to reach other tumors, or metastasized cells in a patient is expected to be, at best, transient due to pre-existing immunity or development of immunity post-injection.

(5) Recombinant HSV vectors (virus particles) capable of systemic tumor homing in HSV-immune patients are desirable. Successful development of this novel type of oHSV will provide a generation of OV's suitable for systemic treatment of metastatic cancer, the central problem in cancer therapy.

SUMMARY

(6) In one aspect of the invention, a retargeted herpes simplex virus (HSV) particle is provided. The virus particle comprises a genome, a capsid, tegument, and an envelope comprising a glycoprotein modified to reduce or eliminate recognition by antibodies that interfere with virus infection by blocking virus attachment and/or entry into a susceptible host cell.

(7) In another aspect, a retargeted herpes simplex virus (HSV) particle is provided. The virus particle comprising an antigenically-modified glycoprotein B (gB) and/or an antigenically-modified glycoprotein D (gD) protein, wherein one or more major epitopes of gB and/or gD reactive with one or more major human serum HSV-neutralizing antibodies is modified to reduce or eliminate binding of the virus particle to a major human serum HSV-neutralizing antibody such that neutralization of the virus by a major human serum HSV-neutralizing antibody is reduced or eliminated.

(8) In another aspect, a recombinant HSV genome, comprising a nucleic acid encoding a retargeted herpes simplex virus (HSV) particle comprising a genome, a capsid, tegument, and an envelope comprising a glycoprotein modified to reduce or eliminate recognition by antibodies that interfere with virus infection by impairing or blocking virus attachment and/or entry into a susceptible host cell. In one embodiment, the virus particle comprises an antigenically-modified glycoprotein B (gB) and/or an antigenically-modified glycoprotein D (gD) protein, wherein one or more major epitopes of gB and/or gD reactive with one or more major human serum HSV-neutralizing antibodies is modified to reduce or eliminate binding of the virus particle to a major human serum HSV-neutralizing antibody such that neutralization of the virus by a major human serum HSV-neutralizing antibody is reduced or eliminated.

(9) In yet another aspect, a viral stock is provided, comprising at least 10⁵ pfus, at least 10⁶ pfus, at least 10⁷ pfus, at least 10⁸ pfus, at least 10⁹ pfus, at least 10¹⁰ pfus, or at least 10¹¹ pfus of retargeted herpes simplex virus (HSV) particles comprising a genome, a capsid, tegument, and an envelope comprising a glycoprotein modified to reduce or eliminate recognition by antibodies that interfere with virus infection by impairing or blocking virus attachment and/or entry into a susceptible host cell. In one embodiment, the virus particles comprise an antigenically-modified glycoprotein B (gB) and/or an antigenically-modified glycoprotein D (gD) protein, wherein one or more major epitopes of gB and/or gD reactive with one or more major human serum HSV-neutralizing antibodies is modified to reduce or eliminate binding of the virus particle to a major human serum HSV-neutralizing antibody such that neutralization of the virus by a major human serum HSV-neutralizing antibody is reduced or eliminated.

(10) A dosage form comprising at least 10⁵ pfus, at least 10⁶ pfus, at least 10⁷ pfus, at least 10⁸ pfus, at least 10⁹ pfus, at least 10¹⁰ pfus, or at least 10¹¹ pfus of

retargeted herpes simplex virus (HSV) particles comprising a genome, a capsid, tegument, and an envelope comprising a glycoprotein modified to reduce or eliminate recognition by antibodies that interfere with virus infection by impairing or blocking virus attachment and/or entry into a susceptible host cell, and a pharmaceutically-acceptable excipient. In one embodiment, the virus particles comprise an antigenically-modified glycoprotein B (gB) and/or an antigenically-modified glycoprotein D (gD) protein, wherein one or more major epitopes of gB and/or gD reactive with one or more major human serum HSV-neutralizing antibodies is modified to reduce or eliminate binding of the virus particle to a major human serum HSV-neutralizing antibody such that neutralization of the virus by a major human serum HSV-neutralizing antibody is reduced or eliminated, and a pharmaceutically-acceptable excipient.

(11) A method of treating a patient having a cancer, comprising administering to the patient an amount of retargeted herpes simplex virus (HSV) particles effective to treat a cancer patient. The retargeted herpes simplex virus (HSV) particles comprise a genome, a capsid, tegument, and an envelope comprising a glycoprotein modified to reduce or eliminate recognition by antibodies that interfere with virus infection by impairing or blocking virus attachment and/or entry into a susceptible host cell, and a pharmaceutically-acceptable excipient. In one embodiment, the virus particles comprise an antigenically-modified glycoprotein B (gB) and/or an antigenically-modified glycoprotein D (gD) protein, wherein one or more major epitopes of gB and/or gD reactive with one or more major human serum HSV-neutralizing antibodies is modified to reduce or eliminate binding of the virus particle to a major human serum HSV-neutralizing antibody such that neutralization of the virus by a major human serum HSV-neutralizing antibody is reduced or eliminated, and a pharmaceutically-acceptable excipient.

(12) The following numbered clauses describe various aspects of the invention Clause 1: A retargeted herpes simplex virus (HSV) particle, comprising a genome, a capsid, tegument, and an envelope comprising a glycoprotein modified to reduce or eliminate recognition by antibodies that interfere with virus infection by impairing or blocking virus attachment and/or entry into a susceptible host cell. Clause 2: The virus particle of clause 1, comprising an antigenically-modified glycoprotein B (gB) and/or an antigenically-modified glycoprotein D (gD) protein, wherein one or more major epitopes of gB and/or gD reactive with one or more major human serum HSV-neutralizing antibodies is modified to reduce or eliminate binding of the virus particle to a major human serum HSV-neutralizing antibody such that neutralization of the virus by a major human serum HSV-neutralizing antibody is reduced or eliminated. Clause 3: The virus particle of clause 2, wherein one or more epitopes of gD are modified to reduce or eliminate binding of the virus particle to a major human serum HSV-neutralizing antibody such that neutralization of the virus by the one or more major human serum HSV-neutralizing antibody is reduced or eliminated. Clause 4: The virus particle of clause 2, wherein one or more epitopes of gB are modified to reduce or eliminate binding of the virus particle to a major human serum HSV-neutralizing antibody such that neutralization of the virus by the one or more major human serum HSV-neutralizing antibody is reduced or eliminated. Clause 5: The virus particle of any one of clauses 1-3, wherein one or more of amino acids 10-20, 54, 75-79, 132, 140, 213, 216, 222-224, and 262-279 of SEQ ID NO: 1 or SEQ ID NO: 2, or one or more amino acids corresponding to amino acids 10-20, 54, 75-79, 132, 140, 213, 216, 222-224, and 262-279 of SEQ ID NO: 1 or SEQ ID NO: 2, are modified in the gD glycoprotein of the virus particle to reduce binding of a major HSV serum neutralizing antibody to the viral particle. Clause 6: The virus particle of clause 5, wherein one or both of amino acids P54 and T213 of SEQ ID NO: 1 or SEQ ID NO: 2, or one or both amino acids corresponding to amino acids P54 and T213 of SEQ ID NO: 1 or SEQ ID NO: 2 are modified, such as P54Q and/or T213M. Clause 7: The virus particle of clause 2, wherein one or more of amino acids 47, 62, 85, 203, 303, 304, 305, 308, 328, 335, 419, 473, 594, or 640-670 of SEQ ID NO: 3 or amino acid 412 of SEQ ID NO: 4, or one or more amino acids corresponding to amino acids 47, 62, 85, 203, 303, 304, 305, 308, 328, 335, 419, 473, 594, or 640-670 of SEQ ID NO: 3 or amino acid 412 of SEQ ID

NO: 4, are modified in the gB glycoprotein of the virus particle to reduce binding of a major HSV serum neutralizing antibody to the viral particle. Clause 8: The virus particle of any one of clauses 1-7, having a $\Delta 38$ mutation of SEQ ID NO: 1 or 2, or a mutation in a gD glycoprotein corresponding to a $\Delta 38$ mutation of SEQ ID NO: 1 or 2. Clause 9: The virus particle of any one of clauses 1-8, comprising a non-native ligand capable of binding a surface component of a target cell type. Clause 10: The virus particle of clause 9, wherein the target cell type is a cancer cell. Clause 11: The virus particle of clause 9, wherein the non-native ligand capable of binding a surface component of a target cell type is incorporated into a viral envelope glycoprotein of the virus particle. Clause 12: The virus particle of clause 11, wherein the viral envelope glycoprotein of the virus particle into which the ligand is incorporated is gD, gC, gB and/or gH. Clause 13: The virus particle of clause 9, wherein the surface component is one or more of EGFR, EGFRvIII, other oncogenic EGFR variants, HER2 00133, CXCR4, carcinoembryonic antigen (CEA), CLC-3/annexin-2/MMP-2, human transferrin receptor, EpCAM, or c-Met. Clause 14: The virus particle of any one of clauses 1-13, wherein binding of at least one or more viral envelope glycoproteins to its natural receptor is eliminated. Clause 15: The virus particle of any one of clauses 1-14, wherein the genome comprises an exogenous expression cassette. Clause 16: The virus particle of clause 15, wherein the exogenous expression cassette encodes an agent that enhances tumor killing activity. Clause 17: The virus particle of any one of clauses 1-16, wherein the genome comprises a target sequence for one or more microRNAs. Clause 18: A recombinant HSV genome encoding the virus particle of any one of clauses 1-17. Clause 19: A viral stock, comprising at least 10⁵ pfus, at least 10⁶ pfus, at least 10⁷ pfus, at least 10⁸ pfus, at least 10⁹ pfus, at least 10¹⁰ pfus, or at least 10¹¹ pfus of a virus particle of any one of clauses 1-17. Clause 20: A dosage form comprising at least 10⁵ pfus, at least 10⁶ pfus, at least 10⁷ pfus, at least 10⁸ pfus, at least 10⁹ pfus, at least 10¹⁰ pfus, or at least 10¹¹ pfus of a virus particle of any one of clauses 1-17, and a pharmaceutically-acceptable excipient. Clause 21: The dosage form of clause 20, provided as a unit dosage form. Clause 22: The dosage form of clause 20, formulated for parenteral delivery. Clause 23: The dosage form of clause 22, formulated for intravenous delivery. Clause 24: A method of treating a patient having a cancer, comprising administering to the patient an amount of the virus particle of any one of clauses 1-17 effective to treat a cancer patient.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A provides aligned exemplary sequences of HSV gD1 and gD2 (SEQ ID NOS: 1 and 2, respectively). Amino acids 2-24 may be replaced with the an scFv sequence, such as the sequence of FIG. 2, or other targeting ligand sequence. FIGS. 1B and 1C provides continuous aligned exemplary sequences of HSV gB1 and gB2 (SEQ ID NOS: 3 and 4, respectively). FIG. 1D provides the sequence of the modified (“retargeted”) gD1, gD:scE $\Delta 38$ (SEQ ID NO: 5), having the anti-EGFR scFv sequence between residues 1 and 25, and a deletion of residue Y38 (438; denoted by a * in the amino acid sequence). The signal peptide (bold text) is shown on a separate line, but is a contiguous sequence with the remainder of the protein prior to its cleavage by signal peptidase.

(2) FIG. 2 provides an exemplary amino acid sequence of an anti-EGFR/EGFRvIII scFv (SEQ ID NO: 6).

(3) FIG. 3 shows schematically the genome structures of wt HSV (a) and KNTc recombinants. (b) KNTc- Δ gD:GW contains bacterial artificial chromosome (BAC) sequences between U.sub.L37 and U.sub.L38 for viral genome propagation and engineering in *E. coli*, a ubiquitin C promoter-mCherry expression cassette (UbC-mCh) between U.sub.L3 and U.sub.L4, two viral entry-enhancing mutations in the gB gene (gB:NT), and a GW recombination cassette in place of the gD

coding sequence (Δ gD:GW) to allow rapid, orientation-specific insertion of reference and modified gD genes under control of the gD promoter. Inserted genes include wt gD1 (gD wt) (c) and retargeted gD (gD:scE Δ 38) (d) without or with P54Q and/or T213M mutations. Known epitope residues for mAbs MC5 and MC23 are indicated above the gD:scE Δ 38 schematic and mutated residues below.

(4) FIG. 4 shows the generation and characterization of B78-EGFRvIII clones (B78-vIII). B78-H1 mouse melanoma cells were infected with an EGFRvIII-expressing recombinant retrovirus. Transduced cells were selected for resistance to 10 μ g/ml blasticidin, pooled and sorted by EGFRvIII expression level using anti-EGFR monoclonal antibody H11 and a fluorescence activated cell sorter (FACS). Cells from the front and back of the EGFRvIII-positive peak were separately cloned by limiting dilution and individual clones were analyzed by Western blot (WB) for EGFRvIII expression. Results show EGFRvIII expression in selected B78-vIII clones. Chinese Hamster Ovary (CHO-K1) cells expressing EGFRvIII (CHO-EGFRvIII) were used as the positive control. B78H1 cells were negative for EGFRvIII (B78H1, far right). B78H1 cells transiently transfected with an EGFRvIII expression plasmid exhibited barely detectable expression of EGFRvIII. β -Actin was visualized as a loading control. B78-vIII clone 11B was chosen for subsequent experimental studies.

(5) FIG. 5. Effects of retargeting and monoclonal antibody (mAb) resistance (mar) mutations on mAb binding to purified gD ectodomains (ECDs). Sequences encoding the ECDs of gD:scE Δ 38 and its mar mutants were cloned into a baculovirus expression plasmid, pVTBac. Sf9 insect cells were transfected with these plasmids along with linearized baculovirus DNA to produce recombinant baculoviruses. Individual isolates were tested by WB for gD ECD production and the highest producer clones were used for larger scale protein production. Soluble proteins were purified through an anti-gD (DL6) column. Binding of 25 gD-specific mAbs to each purified ECD was analyzed by surface plasmon resonance imaging (SPRi) and data for binding of retargeted (A) or mutant retargeted gD (B-D) by each anti-gD mAb are shown as a percentage relative to wt gD (100%). Each mAb is named below the horizontal axis; mAbs to the same or overlapping epitopes are grouped and each group is referred to by a distinct color (Cairns 2017), as indicated below the mAb names. Values are the means $\pm$ SEM of 2-5 independent determinations, except those denoted by (*) that were determined just once. A. gD:scE Δ 38; B. gD:scE Δ 38-P54Q; C. gD:scE Δ 38-T213M; D. gD:scE Δ 38-P54Q/T213M.

(6) FIGS. 6A-6C. gD activity in virus-free fusion assays and inhibition by mAbs. To measure gD fusion activity, an effector cell population was created by transfection of B78H1 cells (no gD receptor) with expression plasmids for gB:NT, gH, gL, different versions of gD, as indicated, and split luciferase N-terminal plasmid RLuc8.sub.(1-7). B78-vIII (EGFRvIII-expressing) target cells were transfected with split luciferase C-terminal plasmid RLuc8.sub.(8-11). Upon mixing of the effector and target cell populations, luminescence resulting from glycoprotein-mediated fusion of target and effector cells was measured over time. (FIG. 6A) Time course of fusion activity relative to gD:scE Δ 38 activity at 6 h (set to 100% luminescence). (FIGS. 6B and 6C) Antibody inhibition was performed by preincubation of the effector cell populations with MC5 (FIG. 6B) or MC23 mAb (FIG. 6C) at the indicated concentrations for 1 h prior to mixing with target cells. Data are shown relative to no antibody at 6 h (100%).

(7) FIGS. 7A and 7B. Virus entry specificities and virion incorporation of gD. (FIG. 7A) Retargeted KNTc viruses produced by BAC transfection of Vero cells were used to infect Vero (nectin-1+/EGFR+), B78-H1, B78/C (nectin-1+), and B78-vIII cells (EGFRvIII+) at MOI=1 for 20 h. Viruses are identified at the top of the columns according to their retargeted gD version. Entry was recorded as mCherry fluorescence. (FIG. 7B) Western blots showing VP16, gD and gB content of equal genome copies of purified virions. Size markers in kDa are indicated at the left for the gD blot.

(8) FIG. 8. Neutralization of retargeted viruses by mAbs MC5 and MC23. Viruses named below

each vertical pair of panels (A, B or C, D) were incubated with virus neutralizing mAbs MC5 (A, C) or MC23 (B, D) at a range of dilutions prior to infection of Vero cells. Infected cell monolayers were overlaid with high-density medium and plaques were counted 48 h later. Representative results showing % plaque-forming units (PFU) relative to virus-only control wells (100%). (A, B) Retargeted and single-mutant retargeted viruses; (C, D) retargeted and double-mutant retargeted viruses. Data represent the means of 3 wells/condition $\pm$ SEM.

(9) FIG. 9—Neutralization of the retargeted and double-mar mutant retargeted viruses. Both viruses were incubated with different mAbs at a range of dilutions prior to infection of Vero cells. Infected cell monolayers were overlaid with high-density medium and plaques were counted 48 h later. Representative results showing % PFU relative to virus-only control wells (100%). Neutralization of the retargeted virus (gD:scE Δ 38) and double-mar mutant retargeted virus (gD:scE Δ 38 P54Q/T213M) with A, mAb H162; B, mAb LP2; C, mAb DL6; and D, mAb MC14. KNTc expressing wt gD is included in panels C and D for comparison (gD wt). Data is shown as the mean of triplicate wells $\pm$ SEM.

DETAILED DESCRIPTION

(10) The use of numerical values in the various ranges specified in this application, unless expressly indicated otherwise, are stated as approximations as though the minimum and maximum values within the stated ranges are both preceded by the word “about”. In this manner, slight variations above and below the stated ranges can be used to achieve substantially the same results as values within the ranges. Also, unless indicated otherwise, the disclosure of these ranges is intended as a continuous range including every value between the minimum and maximum values. For definitions provided herein, those definitions also refer to word forms, cognates and grammatical variants of those words or phrases.

(11) As used herein, the terms “comprising,” “comprise” or “comprised,” and variations thereof, in reference to elements of an item, composition, apparatus, method, process, system, claim etc. are intended to be open-ended, meaning that the item, composition, apparatus, method, process, system, claim etc. includes those elements and other elements can be included and still fall within the scope/definition of the described item, composition, apparatus, method, process, system, claim etc. As used herein, “a” or “an” means one or more. As used herein “another” may mean at least a second or more.

(12) As used herein, the terms “patient” or “subject” refer to members of the animal kingdom, including, but not limited to human beings.

(13) An “effective amount” or “amount effective” to achieve a desirable therapeutic, pharmacological, medicinal, or physiological effect is any amount that achieves the stated purpose. Based on the teachings provided herein, one of ordinary skill can readily ascertain effective amounts of the elements of the described dosage form and produce a safe and effective dosage form and drug product.

(14) In the context of delivery of therapeutic agents, such as the rHSV described herein, delivery profiles may be varied depending on the desired therapeutic effect. The nature of the present disclosure relates to delivery of a recombinant virus particle to cancer cells. As such, the therapeutic agent may preferably be delivered parenterally, such as intravenously. A “dosage form” is the physical form of a dose of a chemical compound used as a drug or medication intended for administration or consumption. A dosage form corresponds to a unit dosage form, even when dispensed from a drug product that includes multiple doses.

(15) The virus particles may be incorporated into an intravenous or other parenteral dosage form which may comprise a pharmaceutically acceptable carrier, or excipient.

Therapeutic/pharmaceutical compositions are prepared in accordance with acceptable pharmaceutical procedures, such as described in *Remington: The Science and Practice of Pharmacy*, 21st edition, ed. Paul Beringer et al., Lippincott, Williams & Wilkins, Baltimore, MD Easton, Pa. (2005) (see, e.g., Chapters 41 and 42 for examples of parenteral and intravenous

formulations and methods of making such formulations). Depending on the delivery route, the dosage form may comprise additional carriers or excipients, such as water, saline (e.g., normal saline), or phosphate-buffered saline, as are broadly-known in the pharmaceutical arts.

Compositions may comprise a pharmaceutically acceptable carrier, or excipient. An excipient is an inactive substance used as a carrier for the active ingredients of a medication. Although “inactive,” excipients may facilitate and aid in increasing the delivery, stability or bioavailability of an active ingredient in a drug product. Non-limiting examples of useful excipients include: anti-adherents, binders, rheology modifiers, coatings, disintegrants, emulsifiers, oils, buffers, salts, acids, bases, fillers, diluents, solvents, flavors, colorants, glidants, lubricants, preservatives, antioxidants, sorbents, vitamins, sweeteners, etc., as are available in the pharmaceutical/compounding arts.

(16) The term “ligand” refers to a binding moiety for a specific target, its binding partner. The molecule can be a cognate receptor, a protein, a small molecule, a hapten, or any other relevant molecule, such as an affibody or a paratope-containing molecule. The term “antibody” refers to an immunoglobulin, derivatives thereof which maintain specific binding ability, and proteins having a binding domain which is homologous or largely homologous to an immunoglobulin binding domain. As such, the antibody operates as a ligand for its cognate antigen, which can be virtually any molecule.

(17) The term “antibody fragment” refers to any derivative of an antibody which is less than full-length. In exemplary embodiments, the antibody fragment retains at least a significant portion of the full-length antibody's specific binding ability. Examples of antibody fragments include, but are not limited to, Fab, Fab', F(ab').sub.2, Fv, Fd, dsFv, scFv, diabody, triabody, tetrabody, di-scFv (dimeric single-chain variable fragment), bi-specific T-cell engager (BiTE), single-domain antibody (sdAb), or antibody binding domain fragments. In the context of targeting ligands, the antibody fragment may be a single chain antibody fragment. Alternatively, the fragment may comprise multiple chains which are linked together, for instance, by disulfide linkages. The fragment may also optionally be a multimolecular complex. A functional antibody fragment will typically comprise at least about 50 amino acids and more typically will comprise at least about 200 amino acids.

(18) Ligands also include other engineered binding reagents, such as affibodies and designed ankyrin repeat proteins (DARPs), that exploit the modular nature of repeat proteins (Forrer T, Stumpp M T, Binz H K, Plückthun A: A novel strategy to design binding molecules harnessing the modular nature of repeat proteins, *FEBS Lett* 2003, 539: 2-6; Gebauer A, Skerra A: Engineered protein scaffolds as next-generation antibody therapeutics, *Curr Opin Chem Biol* 2009, 13:245-255), comprising, often as a single chain, one or more antigen-binding or epitope-binding sequences and at a minimum any other amino acid sequences needed to ensure appropriate specificity, delivery, and stability of the composition (see also, e.g., Nelson, A L, “Antibody Fragments Hope and Hype” (2010) *MAbs* 2(1):77-83).

(19) A virus genome is a nucleic acid that can be packaged in a viral capsid and delivered as part of a virus particle. A viral genome includes any sequences necessary for packaging and delivery of the genome to a target cell for expression of one or more genes encoded by the viral genome. In the context of the HSV virus particles or virions described herein, the genome is packaged in the HSV viral capsid for delivery to target cells.

(20) A significant amount of research has been directed to recombinant, oncolytic HSV viruses (oHSV). Methods of making and using oHSV vectors and viruses are broadly-known (see, e.g., Goins, W F, et al. Retargeting of Herpes Simplex Virus (HSV) Vectors *Curr Opin Virol.* 2016 December; 21: 93-101; Grandi, P, et al. Design and application of oncolytic HSV vectors for glioblastoma therapy. *Expert Rev Neurother.* 2009 April; 9(4): 505-517; and Peters and Rabkin, Designing herpes viruses as oncolytics, *Oncolytics* (2015) 2, 15010, incorporated herein by reference for their technical disclosure). Likewise, the overall structure of various HSV components are well-studied. In 2015, an oHSV (T-VEC, IMLYGIC™, talimogene laherparepvec)

became the first oHSV to receive FDA approval based on its proven safety and efficacy in the treatment of melanoma patients. The effectiveness of this virus has relied in part on vector arming with granulocyte macrophage colony stimulating factor (GM-CSF) to enhance the recruitment of antigen presenting cells. However, as indicated above, implementation has, in part, been hampered by development of immunity to HSV, e.g. to HSV glycoproteins gB and gD. Following intra-tumor delivery of T-VEC for treatment of melanoma tumors, there was no correlation between anti-HSV antibody titers and therapeutic responses, but all seronegative patients seroconverted within 3-4 weeks. A large fraction of the human population is HSV seropositive and systemic treatment of metastatic cancer with even tumor-targeted HSV will likely be impaired in these patients. As such, a “stealthed” HSV, e.g., an oHSV that is concealed from pre-existing immunity, is described herein. (21) Amino acid sequences of exemplary HSV gB and gD glycoproteins are provided in FIGS. 1A-1C (SEQ ID NOS: 1-4). A number of studies have produced detailed epitope maps for of HSV gD and gB (see, e.g., Atanasiu D, et al. 2018. Using Antibodies and Mutants to Localize the Presumptive gH/gL Binding Site on Herpes Simplex Virus gD. *J Virol* 92:e01694-18; Cairns, T. M., et al. 2015. *J Virol* 89:9213-9231; Bender F C, et al. 2007. Antigenic and mutational analyses of herpes simplex virus glycoprotein B reveal four functional regions. *J Virol* 81(8):3827-3841; and Cairns T M, et al. 2006. Epitope Mapping of Herpes Simplex Virus Type 2 gH/gL Defines Distinct Antigenic Sites, Including Some Associated with Biological Function. *J Virol* 80:2596-2608; and Cairns T M, et al. (2017) Global Sensing of the Antigenic Structure of Herpes Simplex Virus gD Using High-Throughput Array-Based SPR Imaging. *PLoS Pathog* 13(6):e1006430 (Cairns et al. 2017)), including a more detailed understanding of the function of major serum antibody binding sites, including the relationship of the binding sites to viral neutralization, receptor binding, viral entry, and membrane fusion.

(22) A “neutralizing antibody” is an antibody or fragment thereof that binds a virion and reduces the infectivity of that virion. The ability of an antibody to neutralize a virus can be effectively tested, for example and without limitation, by a plaque assay as is broadly-known, e.g., in Cairns et al. 2017. In the context of the present invention, “major” neutralizing antibodies are neutralizing antibodies present in a majority of the population, such as in at least 10%, at least 20%, at least 30%, at least 40%, or at least 50% of a human population seropositive for HSV, e.g., HSV-1 or HSV-2, and selective for a specific epitope, e.g., epitopes described herein, and in Cairns et al. 2017 as MC5, MC23, DL11, MC14,1D3, and 11 B3AG, or as “red community”, “pink community”, or “blue community” epitopes, or in groups Ia, Ib, and III, as shown in Cairns et al. 2017 (reproduced in table 1, below) (see also, Cairns, T M, et al. (2015) Patient-Specific Neutralizing Antibody Responses to Herpes Simplex Virus Are Attributed to Epitopes on gD, gB, or Both and Can Be Type Specific. *J. Virology* 89(18):9213-9231 (“Cairns et al. 2015”)). For gB, major epitopes include SS144, C226, and SS10, as indicated in Cairns et al. 2015 (see, also, Cairns et al. 2014. Mechanism of Neutralization of Herpes Simplex Virus by Antibodies Directed at the Fusion Domain of Glycoprotein B. *J. Virol.* 88(5):2677-2689; Bender F C, et al. 2007. Antigenic and mutational analyses of herpes simplex virus glycoprotein B reveal four functional regions. *J Virol* 81(8):3827-3841; Kousoulas, K G, et al., (1988) Antibody-resistant mutations in cross-reactive and type-specific epitopes of herpes simplex virus 1 glycoprotein B map in separate domains *Virology* 166(2):423-431; Sanchez-Pescador, L, et al. (1993) Antibodies to Epitopes of Herpes Simplex Virus Type 1 Glycoprotein B (gB) in Human Sera: Analysis of Functional gB Epitopes Defined by Inhibition of Murine Monoclonal Antibodies *J. Infectious Diseases* 168(4):844-853; and Highlander, S L, et al. (1989) Identification of mar Mutations in Herpes Simplex Virus Type 1 Glycoprotein B Which Alter Antigenic Structure and Function in Virus Penetration *J. Virol* 63(2):730-738). Neutralizing antibodies may be classified with respect to their target epitope in a virion, such as an envelope glycoprotein, e.g., gD or gB. Major neutralizing antibodies bind to a corresponding “major epitope” in an envelope glycoprotein.

(23) TABLE-US-00001 TABLE 1 Properties of anti-gD mAbs (Cairns et al. 2017). Plaque Assay

IgG ($\mu\text{g/mL}$) for 50% gD neutralization of: Group MAb binding Epitope residues HSV-1 (KOS) HSV-2 (333) Ia HD1 TC 216 4 $\pm$ 1.4 4.7 $\pm$ 1.8 HD2 TC ND 4 $\pm$ 1.4 6.5 $\pm$ 2.1 LP2 TC 216 2.6 $\pm$ 3.3 2.7 $\pm$ 1.1 MC23 TC 213, 216 1.6.sup.a 0.26.sup.a DL15* TC ND 3.2 $\pm$ 4 NN Ib DL11 TC 36, 132, 140, 222-224 0.004.sup.a 0.31.sup.a 77S* TC 36, 222-224 0.1 $\pm$ 0.1 1.4 $\pm$ 1.6 97S* TC 36, 222-224 0.9 $\pm$ 0.1 1.8 $\pm$ 2.3 106S* TC 38, 222-224 2.3 $\pm$ 1 2.1 $\pm$ 1.9 108S* TC 38, 222-224 0.6 $\pm$ 0.5 0.4 $\pm$ 0.5 IIa MC4 TC 262-272 NN.sup.a NN.sup.a MC8 TC 262-272 NN.sup.a NN.sup.a MC9 TC 262-272 NN.sup.a NN.sup.a MC10 TC 262-272 NN.sup.a NN.sup.a MC14 TC 262-272 NN.sup.a NN.sup.a MC15 TC 262-272 NN.sup.a NN.sup.a BD78 TC 262-272 21 $\pm$ 12.7 12 $\pm$ 4.2 BD80 TC 262-272 19 $\pm$ 6.6 9 $\pm$ 0 IIb DL6 TC 272-279 NN.sup.a NN.sup.a IIc 4E3E* TC ND 2.3 $\pm$ 2.4 5 $\pm$ 5.6 4G4D* TC ND 3 $\pm$ 0 16.2 $\pm$ 18 III VID TC 54 3.sup.b 13.sup.b 11S TC ND 5.3 $\pm$ 2.6 NN 3D5* T1S ND 6.7 $\pm$ 1.1 NN MC5 TC 75-79 3.1.sup.a 6.2.sup.a H162* TC ND 20.7 $\pm$ 27.2 7.3 $\pm$ 5.1 H193* TC ND 21.7 $\pm$ 25.8 39 $\pm$ 93 IV 45S T1S ND 21.3 $\pm$ 17.2 NN D10-G12* TC ND 2.5 $\pm$ 1.9 NN VII 110S TC 1-29 1.8 $\pm$ 2.4 5 $\pm$ 5.6 1D3 TC 10-20 0.39.sup.a 6.2.sup.a MC1 TC 10-20 NN NN H170 TC 1-23 5.9 $\pm$ 1.6 20.7 $\pm$ 14.1 X HD3 TC ND 4 $\pm$ 0.7 22.5 $\pm$ 10.6 XII AP7 TC 25, 27, 294 24.7 $\pm$ 22.7 NN 12S* TC ND 2.5 $\pm$ 1.4 NN XVII 11B3AG* TC ND 0.9 $\pm$ 0.6 NN A18* T1S 246 12.5 $\pm$ 7.8 NN MG2 T2S 248 NN.sup.a 0.78.sup.a

(24) It should be recognized that the genome sequences of several HSV strains are known to persons of ordinary skill (e.g., MacDonald, *J. Virol.*, 86(11): 6371 (2012); McGeoch, *J. Gen. Virol.*, 69: 1531-1574 (1988); GenBank Accession No. J0673480; NCBI Reference Sequence: NC_001806.1; MacDonald, *J. Virol.* 86(17): 9540 (2012); GenBank Accession No. JX142173, which are incorporated herein by reference). Accordingly, manipulation of the sequence of HSV genes and loci is within the level of ordinary skill. It should also be noted that these published sequences, and the sequences provided herein (SEQ ID NOS: 1-4), are merely exemplary and that other strains or variants of HSV can be employed as a source genome in engineering the inventive virus particle or vector.

(25) An HSV particle may be retargeted by disrupting, e.g. deleting or otherwise rendering ineffective, targeting of the virion to its normal binding targets, including nectin-1 (CD111/HveC), nectin-2 (CD112/HveB), HVEM (TNFRSF14/HveA), and 3-O-sulfated heparan sulfate (3-OS HS). Nectin-1 is the main receptor for HSV-1 and HSV2, nectin-2 is a receptor for HSV-2 and for certain mutant strains of HSV-1 (ANG, KOS-rid1, KOS-rid2) that have a mutation that blocks HVEM binding (e.g. Q27P). HVEM is mainly expressed on lymphocytes. 3-OS HS is a less well-defined receptor for HSV-1 gD only whose binding can be eliminated by mutations in the HVEM-binding region (Yoon et al, DOI: 10.1128/JVI.77.17.9221-9231.2003). In the case of gD of HSV-1, deletion of Y38, and replacement of the HVEM targeting moiety with a retargeting ligand sequence, such as an scFv, or another ligand, such as a single-domain (VHH) antibody, an affibody or a natural receptor binding partner sequence, such as a growth factor, as are broadly-known in the recombinant HSV field (see, e.g., U.S. Pat. No. 9,593,347 and Goins, W F, et al. *Curr Opin Virol.* 2016 December; 21: 93-101), can be used to alter the specific receptor dependence of HSV infection, i.e. retarget the virus.

(26) As used herein, "modification" of an amino acid sequence, protein, epitope, antigenic determinant, etc. includes any mutation type that changes the amino acid sequence of a protein. In embodiments, the modifications are non-conservative in that they alter the binding of one or more antibodies to the protein. Suitable modifications include, without limitation: deletion of one or more amino acids, substitution of one or more stated amino acids (e.g., missense mutations), or insertion of one or more amino acids. Although specific amino acid substitutions are listed in the examples below (for HSV-1 gD (gD1), P54Q and T213M), substitution of the same amino acids, that is, amino acid residues P54 and/or T213 of gD1, with certain other amino acids may modify the relevant epitope, e.g. antibody binding, preferably without substantially disrupting infectivity of the virion for a target cell. Likewise, other amino acids associated with epitopes of HSV-1 gD (gD1

or HSV-2 gD (gD2)), such as amino acids 10-20, 75-79, 132, 140, 213, 216, 222-224, and 262-279 of SEQ ID NOS: 1 or 2 or amino acids corresponding to those amino acids, amino acids associated with major epitopes found in HSV-1 gB (gB1) or HSV-2 gB (gB2), such as, referencing SEQ ID NO: 3, amino acids 47 (e.g., N47T), 62 (e.g., A62T), 85 (e.g., G85D), 203 (e.g., A203T), 303 (e.g., Y303N), 304 (e.g., R304Q), 305 (e.g., E305K), 308 (e.g., H308Y), 328 (e.g., R328H), 335 (e.g., R335Q), 419 (D419K), 473 (e.g., S473N), 594 (e.g., G594R), 640-670, or, referencing SEQ ID NO: 4, amino acid 412 (e.g., E414K), or amino acids corresponding to those amino acids, and other amino acids in major epitopes of gD1, gD2, gB1, gB2, or retargeted versions thereof, corresponding to neutralizing antibodies, may be mutated to effectively modify the epitope in a manner consistent with the object of reducing neutralization, yet maintaining infectivity in a target cell. Further, determining the impact on antibody binding of any modification in any given epitope, and whether or not such modification can negatively impact infectivity in a target cell, is readily tested, such as in a plaque assay, western blotting or antibody binding assay, e.g. as described herein.

(27) Many versions of HSV gD and gB that may find use, in original or modified form, in the virus particles described herein, are known to those of ordinary skill in the art. Those versions of gD and gB may have the wild-type (wt) sequence, e.g., as recited in SEQ ID NOS: 1-4, or variants thereof, including naturally-occurring or man-made variants thereof, or precursors thereof. Further, the numbering of the amino acids in HSV gD and gB glycoproteins may be presented differently, depending on the source of the sequence information. As such, where specific amino acids are referenced herein, they are in reference to amino acids in the context of SEQ ID NOS: 1-4. Unless specifically referring to the sequences of SEQ ID NOS: 1-4, modifications in amino acid sequences are therefore said to “correspond to” the referenced amino acids in SEQ ID NOS: 1-4, meaning the equivalent amino acids are modified in gD or gB glycoproteins that may vary in sequence or sequence numbering from the sequences of SEQ ID NOS: 1-4.

(28) The present invention provides a recombinant HSV (rHSV), such as an oHSV, comprising a modified gB and/or a modified gD envelope glycoprotein in which one or more epitopes representing binding sites for one or more serum antibodies in HSV seropositive individuals are modified to reduce or eliminate neutralization of the recombinant HSV when administered to a patient seropositive for the one or more serum antibodies. That is, antibody binding of one or more major serum antibodies to gD or gB is reduced or eliminated. It is noted that mAbs are used as proxies for major serum antibodies in the examples.

(29) By “recombinant”, it is meant the viral genome is genetically modified or genetically engineered to encode amino acid changes, such as insertions, deletions, or substitutions, not present in the wild-type (wt) genome. As such, a protein sequence is considered to be “modified” if it has a sequence different from a wt (e.g., reference) sequence, such as one of SEQ ID NOS: 1-4. The rHSV may be derived from or engineered from HSV-1 acid/or HSV-2 genomes, including genomes of any suitable strain(s) or genetically-engineered version(s) of those genomes.

(30) In practice, a recombinant HSV can be detargeted, and optionally retargeted. The recombinant HSV is detargeted or retargeted by removal of or destruction of the native cell-surface binding moieties in gD, e.g. the nectin-1 binding site by removal of, or substitution of Y38 (e.g., Δ 38), and the HVEM binding sequence by removal or destruction, and optionally replacing the HVEM binding sequence with a ligand for binding a cell surface binding partner present on a target cell, such as an scFv sequence, for example and without limitation encoding an anti-EGFR/EGFRvIII scFv (see, e.g. FIG. 2), for targeting human glioblastoma cells, and other cancer cells. Optionally a cell-targeting ligand may be expressed as a separate protein or part of an envelope glycoprotein other than gD, such as in gB or gH (see, e.g., Petrovic, B, et al. Dual Ligand Insertion in gB and gD of Oncolytic Herpes Simplex Viruses for Retargeting to a Producer Vero Cell Line and to Cancer Cells. *J. Virology* February 2018, 92 (6) e02122-17; Petrovic B, et al. (2017) Insertion of a ligand to HER2 in gB retargets HSV tropism and obviates the need for activation of the other entry

glycoproteins. *PLoS Pathog* 13(4): e1006352; and Gatta V, et al. (2015) The Engineering of a Novel Ligand in gH Confers to HSV an Expanded Tropism Independent of gD Activation by Its Receptors. *PLOS Pathogens* 11(5): e1004907). Thus, by “retargeted”, it is meant that a virus, e.g. an HSV virus is genetically or synthetically altered such that an envelope glycoprotein is modified to eliminate recognition of the normal virus cognate receptors (referred to as “detargeting”), and provided with a ligand, or mutation, to enable virus attachment and entry through recognition of a cell surface structure other than the normal virus cognate receptors.

(31) The viral genome also may comprise one or more microRNA target sequences to reduce or prevent off-target infection and replication of the viral genome. Also provided herein are virus stocks and pharmaceutical compositions comprising the described rHSV, as well as methods for killing tumor cells employing the described rHSV (oHSV).

(32) As described above, the HSV particle may comprise a non-HSV ligand specific for a molecule (protein, lipid, or carbohydrate determinant) present on the surface of a cell (such as a cancer cell) and/or one or more copies of one or more microRNA target sequences inserted into one or more HSV gene loci, such as one or more HSV gene(s) required for replication of HSV in normal cells.

(33) The non-HSV ligand of the recombinant HSV (rHSV) is incorporated into a glycoprotein exposed on the HSV surface, such as gD or gC, to facilitate targeting the desired cell with the ligand. For example, the ligand can be incorporated between residues 1 and 25, between residues 24 and 25, or between residues 7 and 39 of gD. Ligands for targeting cancer cells, such as glioblastoma or breast cancer, include those targeting EGFR and EGFRvIII, HER2 (human epidermal growth factor receptor 2), CD133, CXCR4 (chemokine receptor type 4, fusin), carcinoembryonic antigen (CEA), CLC-3/annexin-2/MMP-2 (chlorotoxin receptor), human transferrin receptor, epithelial cellular adhesion molecule (EpCAM), or c-Met, and the ligand can target such a receptor or cell-surface molecule, i.e., the ligand is capable of specifically binding such receptor or cell-surface molecule. EGFR- and EGFRvIII-specific ligands, such as antibodies, scFvs (single chain antibodies) and VHHs (single domain antibodies), have been described in the literature (Kuan et al, *Int J. Cancer*, 88, 962-69 (2000); Wickstrand et al., *Cancer Res.*, 55(14):3140-8 (1995); Omidfar et al., *Tumor Biology* 25:296-305 (2004); see also Uchida et al, *Molecular Therapy*, 21:561-9 (2013); see also Braidwood et al., *Gene Ther.*, 15, 1579-92 (2008)).

(34) The rHSV also or alternatively can be targeted by incorporating ligands to other cell-surface molecules or receptors that are not necessarily cancer-associated. For example, ligands can include binding domains from natural ligands (e.g., growth factors (such as EGF, which can target EGFR, NGF, which can target trkA, GDNF, which can target GFR α 1, and the like)), peptide or non-peptide hormones, peptides selected for binding a target molecule (e.g., DARPins and affibodies), etc. The rHSV also can include a mutant form of gB and/or gH that facilitates virus entry through non-canonical receptors (and would typically also have such mutations in one or both of these genes within the rHSV genome).

(35) An exemplary microRNA target sequence for inclusion in the rHSV vector (for example as multiple copies thereof in tandem) is a binding sequence for miR-124, which has particular application for neural applications (e.g., to protect non-cancerous neurons when employing the rHSV for treating nervous-system tumors, such as GBM). Other microRNA target sequences can alternatively be employed for protecting other types of tissues, and it is within the ordinary skill in the art to select a suitable microRNA target sequence to protect a desired tissue or cell type. For example, miR-122 and miR-199 are expressed in normal liver cells but not primary liver cancer; thus one, or a combination of miR-122 and/or miR-199 microRNA target sequences can be employed in an embodiment of the rHSV for treatment of liver cancers. Similarly, target sequences for miR-128 and/or miR-137 microRNA can be employed in oHSV for protection of normal brain. An exemplary microRNA target sequence can be the reverse complement of the microRNA. Where the rHSV is administered systemically, one or more microRNA target sequences are employed to protect non-target cells throughout the body.

(36) The microRNA target sequence(s) may be included in the 3' untranslated region ("UTR") of an HSV gene, to silence that gene in the presence of the microRNA. It may be preferred that multiple copies (such as two copies, three copies, four copies, five copies, six copies, or more) of the microRNA target sequence are inserted in tandem. The multiple copies of the micro-RNA target sequence may be separated by spacers of four or more nucleotides (more preferably eight or more nucleotides). Multiple copies of the microRNA target sequences may be the same or different, such that target sequences of two or more microRNAs are presented in a single gene.

(37) In embodiments of the rHSV, to assist in protecting non-cancerous cells from the lytic effect of HSV infection, multiple copies of one or more microRNA target sequence are inserted in the 3' UTR of an HSV gene that is essential for replication in non-cancerous cells, which are known to persons of ordinary skill. In examples, the site is the 3' UTR of the microRNA-targeted gene in its normal (or native) locus within the HSV genome. In one embodiment, a plurality of microRNA target sequences are inserted into the 3' UTR of the ICP4 gene, such as both copies of the ICP4 gene, in vectors which have both native copies of the ICP4 gene.

(38) The genome of the rHSV vector additionally can comprise one or more exogenous expression cassettes containing encoding sequences, e.g., an open-reading frame (ORF), in operable linkage with promoters, enhancers, and other suitable regulatory elements for expression of the ORF. The encoding sequences, e.g., ORF, encode a gene product such as a protein, such as a reporter protein (such as green fluorescent protein), an oncolytic factor or agent that enhances tumor killing activity (such as tumor necrosis factor ("TNF") or TNF-related apoptosis-inducing ligand ("TRAIL"), or other therapeutically-important gene product (e.g., peptides, drug-activating enzymes, antibodies, therapeutic or regulatory RNAs, and the like).

(39) Exemplary exogenous expression cassettes encode proteins or polypeptides that induce patient immune responses against the cancer or tumor to which the inventive HSV is to be employed to treat. For example, such expression cassettes can include one or more nucleic acids encoding factors such as cytokines (e.g., IL-12 and IFN- β), an antibody directed against cytotoxic T-lymphocyte-associated protein 4 ("CTLA-4") (Hodi et al., *N. Engl. J. Med.*, 363(8): 711-23 (2010)), an antibody directed against either the ligand of programmed cell death protein 1 ("PD1") or the receptor itself (Topalian et al., *N. Engl. J. Med.*, 366(26): 2443-54 (2012)), or epithelial cell adhesion molecule ("EpCAM") (Patriarca et al., *Cancer Treatment Rev*, 38: 68-75 (2012)). EpCAM can serve both as cellular target and rHSV targeting ligand. In one embodiment, an exogenous expression cassette encodes granulocyte-macrophage colony-stimulating factor ("GM-CSF"). The cassette can also encode proteins for "rewiring of aberrant signaling to effector release" (RASER), e.g., as described in Chung, H K, et al, (2019). A compact synthetic pathway rewires cancer signaling to therapeutic effector release. *Science* 364(6439), eaat6982, DOI: 10.1126/science.aat6982.

(40) Other expression cassettes encode proteins or polypeptides that catalyze the conversion of prodrugs to active agents. For example, such expression cassettes can encode enzymes such as cytosine deaminase, which can convert 5-fluorocytosine ("5-FC") into 5-fluorouracil ("5-FU") locally in tumors or cancerous cells infected with the inventive vector (see, e.g., Akimoto et al., *J. Ophthalmol.*, 86(5): 581-86 (2002)), so as to permit 5-FU to act locally within such cells or tumors while minimizing systemic exposure to 5-FU. Similarly, such an expression cassette can encode thymidine kinase (tk) (e.g., operably linked to an HSV immediate-early promoter or strong constitutive promoter), which can activate ganciclovir, or purine nucleoside phosphorylase (PNP), which can block or attenuate the activity of ribonucleotide reductase. In certain embodiments, the inventive vectors also can contain a functional native HSV tk gene.

(41) Within the rHSV vector described herein, the encoding sequences within the exogenous expression cassettes can be in operable linkage with any desired genetic regulatory sequence, such as constitutive promoters or inducible or tissue-specific promoters, many examples of which are known in the art, or microRNA target sites. For example, a commonly-employed constitutive

promoter is the human cytomegalovirus (hCMV) promoter, and other promoters also can be used, e.g., the CMV early enhancer/chicken beta actin (CAG) promoter, the HSV immediate early promoter (e.g., ICP4 promoter), and the like.

(42) Also, in certain embodiments, the genome of the inventive vector contains a deletion of the internal repeat (joint) region comprising one copy each of the diploid genes ICP0, ICP34.5, LAT and ICP4 along with the promoter for the ICP47 or ICP22 gene. In other embodiments, instead of deleting the joint, the expression of genes in or flanking the joint region, particularly ICP0, ICP34.5, and/or ICP47, can be silenced by deleting these genes or otherwise subjecting them to limited mutagenesis impeding their expression or product functionality.

(43) The rHSV vector can be produced by standard methods known to persons of ordinary skill in the field of HSV virology. However, to facilitate manipulation of the HSV genome and production of the inventive vector, the invention also provides a nucleic acid representing the inventive vector genome. In one aspect, the nucleic acid is a bacterial artificial chromosome ("BAC") encoding the rHSV vector, which facilitates manipulation of the HSV in a bacterial system.

(44) The rHSV described herein can be used to target and kill cancerous cells, whether in vivo or in vitro. In one application, the rHSV vector is employed therapeutically, e.g., in human patients and/or against human tumors/cells (which can be xenografts in various mammalian species). However, the method also may be employed in animals, such as companion animals (e.g., cats and dogs), or animals of agricultural importance (e.g., cattle, sheep, horses, and the like), or of zoological importance. Exemplary tumors/cancerous cells, the treatment of which the inventive vectors can be employed for, include: cancers of the central nervous system, such as glioblastoma multiforme (e.g., EGFR/vIII-targeted), breast cancers (e.g., HER2-targeted), squamous cell carcinoma of the head and neck (HNSCC, e.g., EGFR/vIII-targeted), prostate cancers (e.g., EGFR/vIII-targeted), and lung cancers (NSCLC) (e.g., EGFR/vIII-targeted).

(45) Generally, the rHSV vector is most useful when enough of the virus can be delivered to a cell population to ensure that the cells are confronted with a suitable number of virus particles. Thus, the present invention provides a stock, e.g. a homogeneous stock, comprising the rHSV vector. The preparation and analysis of HSV stocks is well known in the art. For example, a viral stock can be manufactured in roller bottles or multi-layer cell stacks containing cells transduced with the rHSV vector. The viral stock can then be purified on a continuous density gradient medium, such as a NYCODENZ® gradient, and aliquotted and stored until needed. Viral stocks vary considerably in titer, depending largely on viral genotype and the protocol and cell lines used to prepare them. It may be preferred that such a stock has a viral titer of at least about $10^{5.5}$ plaque-forming units (pfu), such as at least about $10^{6.6}$ pfu, at least about $10^{7.7}$ pfu, at least about $10^{8.8}$ pfu, at least about $10^{9.9}$ pfu, at least about $10^{10.10}$ pfu, or at least about $10^{11.11}$ pfu.

(46) The invention additionally provides a composition comprising the inventive oHSV vector and a carrier, preferably a physiologically-acceptable or pharmaceutically-acceptable carrier. The carrier of the composition can be any suitable carrier for the vector. The carrier typically will be liquid, but also can be solid, or a combination of liquid and solid components. The carrier desirably is a pharmaceutically acceptable (e.g., a physiologically or pharmacologically acceptable) carrier (e.g., excipient or diluent). Pharmaceutically acceptable carriers are well-known and are readily-available. The choice of carrier will be determined, at least in part, by the particular vector and the particular method used to administer the composition. The composition can further comprise any other suitable components, especially for enhancing the stability of the composition and/or its end-use. Accordingly, there is a wide variety of suitable formulations of the composition of the invention. The following formulations and methods are merely exemplary and are in no way limiting.

(47) Formulations suitable for parenteral administration include aqueous and non-aqueous, isotonic sterile injection solutions, which can contain anti-oxidants, buffers, bacteriostats, and solutes that render the formulation isotonic with the blood of the intended recipient, and aqueous and non-

aqueous sterile suspensions that can include suspending agents, solubilizers, thickening agents, stabilizers, and preservatives. The formulations can be presented in unit-dose or multi-dose sealed containers, such as ampules and vials, and can be stored in a freeze-dried (lyophilized) condition requiring only the addition of a sterile liquid excipient, for example, water, for injections, immediately prior to use. Extemporaneous injection solutions and suspensions can be prepared from sterile powders, granules, and tablets of the kind previously described.

(48) In addition, the composition may comprise additional therapeutic or biologically-active agents, or the additional therapeutic or biologically-active agents may be co-administered with the rHSV. For example, therapeutic factors useful in the treatment of a particular indication can be present or co-administered. Factors that control inflammation, such as ibuprofen or steroids, can be part of the composition or co-administered to reduce swelling and inflammation associated with in vivo administration of the vector and physiological distress. Immune system suppressors can be administered with the composition method to reduce any immune response to the vector itself or associated with a disorder. Alternatively, immune enhancers can be included in the composition or co-administered to upregulate the body's natural defenses against disease, particularly against the cancer or tumor against which the inventive vector is to be used. Antibiotics, e.g., microbicides and/or fungicides, can be included in the composition or co-administered to reduce the risk of infection associated with gene transfer procedures and other disorders.

(49) A retargeted rHSV was prepared as depicted in FIG. 3. The HSV-1 genome consists of two unique regions (unique long [UL], 108 kb, and unique short [US], 13 kb), each flanked by large inverted repeats. A recombination cassette (GW) was inserted in place of the gD coding sequence (Δ gD:GW) to allow rapid, orientation-specific insertion of altered gD genes under control of the gD promoter. This backbone contains bacterial artificial chromosome (BAC) sequences that allow generation of mutant viral genomes in bacteria and production of infectious viruses by transfection of suitable mammalian cell lines. The construct further contains a ubiquitin C promoter (UbCp) driven mCherry reporter gene to observe retargeted virus entry and a viral entry-enhancing mutant gB gene (gB:NT or gB:N/T; Uchida et al, A Double Mutation in Glycoprotein gB Compensates for Ineffective gD-Dependent Initiation of Herpes Simplex Virus Type 1 Infection. *J Virol* 2010: 12200-12209. DOI: 10.1128/JVI.01633-10; Uchida et al, Effective treatment of an orthotopic xenograft model of human glioblastoma using an EGFR-retargeted oncolytic herpes simplex virus. *Mol Ther* 21, 561-569, doi:10.1038/mt.2012.211 (2013)). gD wt includes the wild-type gD sequence from strain KOS, including the signal peptide (residues -25 to -1), HVEM binding site, nectin-1 binding site (including Y38), the transmembrane region (TM), and the cytoplasmic tail region (CT). The open reading frame of gD:scE Δ 38 deletes residue Y38, disrupting the nectin-1 binding site, and replaces residues 2-24 of the HVEM binding site with anti-EGFR/EGFRvIII scFv sequence—effectively retargeting the rHSV (FIGS. 7A and 7B).

EXAMPLES

(50) Approximately 70% of the human population is HSV seropositive, Therefore, we asked whether changing the epitope structure of retargeted gD by mutagenesis of epitopes recognized by HSV-neutralizing anti-gD antibodies would reduce retargeted gD and retargeted virus recognition by monoclonal, and possibly polyclonal, anti-gD antibodies to increase virus resistance to virus-neutralizing (VN) sera. We compared the binding of a panel of monoclonal antibodies (mAbs) that mimic antibody specificities in human HSV-immune sera to the purified ectodomains of wild-type and retargeted gD, revealing the retention of two prominent wild-type gD epitopes in retargeted gD. Substitution of a key residue in each epitope, separately and together, led to the following observations: Both substitutions (i) blocked retargeted gD recognition by mAbs to the respective epitopes, and in combination, caused a global reduction in mAb binding; (ii) protected against fusion inhibition by VN mAbs reactive with each epitope in virus-free cell-cell fusion assays; and (iii) increased the resistance of retargeted HSV-1 to these VN mAbs. Mutagenesis to eliminate additional shared epitopes is expected to further decrease retargeted virus susceptibility to

neutralizing antibodies

(51) Materials and Methods:

(52) Plasmids

(53) pENTR-based plasmids for viral genome modification: The complete gD wt and gD:scEΔ38 coding sequences (e.g., encoding SEQ ID NO: 1) were cloned between the Gateway (GW)-compatible attL sites of plasmid pENTR1A. The P54Q substitution was generated by introducing an internal BstBI site, to insert a synthetic DNA fragment (GenScript) specifying the codon 54 change from CCG to CAG flanked by BstBI and BspEI sites. Likewise, the T213M substitution was introduced by replacement of a synthetic DNA fragment (GenScript) specifying the codon T213M change from ACG to ATG with the use of KasI and FspI sites.

(54) pVT-Bac-based plasmids for gD ectodomain expression and purification: The coding sequences for the mature ectodomains (residues 1-306) of wt, retargeted and mutant retargeted gDs were isolated by PCR amplification with primers pVT Bac gD:scEGFR Δ38 5',

CCAGCCCGGGCAAAGACATTCTAATGACCCAATCTC (SEQ ID NO: 7), introducing a SmaI site (underlined); and pVT Bac gD:scEGFR Δ38 3'

GGTATGCGGCGCGCTTAATGGTAAGGCGTCGCGGCGTCCT (SEQ ID NO: 8), introducing a NotI site (underlined). Respective pENTR recombinants were used as templates and the PCR products were cloned into baculovirus expression plasmid pVT-Bac to produce soluble proteins.

(55) pcDNA3.1-based plasmids for gD expression in mammalian cells: The coding sequences for wt and retargeted gDs were cloned into a modified pcDNA 3.1 plasmid that contains a Gateway (GW) recombination cassette between the CMV promoter and bovine growth hormone (bGH) polyadenylation region (pcDNA-GW; Reinhart et al., An HSV-based library screen identifies PP1α as a negative TRPV1 regulator with analgesic activity in models of pain, *Mol Ther* 2016, 3:16040, doi:10.1038/mtm.2016.40) by LR Clonase II (ThermoFisher, Waltham, MA)-mediated recombination with the respective pENTR-based plasmids.

(56) Other plasmids: gB:NT expression plasmid pCagB:NT was as described in Uchida, H., et al., Novel mutations in gB and gH circumvent the requirement for known gD Receptors in herpes simplex virus 1 entry and cell-to-cell spread. *J Virol*, 2013. 87(3):p. 1430-42. gH and gL expression plasmids pPEP100 and pPEP101, respectively, were kindly provided by Patricia Spear (Northwestern University). PT3.5/CMV-EGFRvIII, made available by John Ohlfest (University of Minnesota), was purchased from Addgene (plasmid #20280). pCX4-bsr-DEST, pCL-gag-pol and pHCMV-VSVG were kind gifts from Akihiro Umezawa (NRICH, Tokyo, Japan). pCX4-bsr-EGFRvIII was constructed by PCR amplification of the EGFRvIII coding sequence from plasmid PT3.5/CMV-EGFRvIII (Weisner, S M, et al. De novo induction of genetically engineered brain tumors in mice using plasmid DNA. *Cancer Res.* 2009 Jan. 15; 69(2):431-9. doi: 10.1158/0008-5472.CAN-08-1800) with primers SalI-EGFRvIIIF (gactagtcgacAATTCGTTGGCCGCATGCGA, SEQ ID NO: 9) and XhoI-EGFRvIIIR (cactactcgagTCATGCTCCAATAAATTCAGTCTTTG, SEQ ID NO: 10). The PCR fragment was transferred into pENTR1a (ThermoFisher) by SalI-XhoI enzyme digestion and cloning between the SalI and XhoI sites of pENTR1a. The EGFRvIII coding sequence was transferred from pENTR1a to pCX4-bsr-DEST by LR Clonase II-mediated recombination. All new constructs were confirmed by DNA sequencing.

(57) Viruses A GW-compatible gD-null viral backbone, KNTc-ΔgD:GW (FIG. 3), on a bacterial artificial chromosome (BAC) was derived from KNTc BAC (Miyagawa, Y., et al., Herpes simplex viral-vector design for efficient transduction of nonneuronal cells without cytotoxicity. *Proc Natl Acad Sci USA*, 2015. 112(13): p. E1632-41) by Red-mediated replacement of the gD coding sequence with a GW cassette, GW-Zeo, amplified with primers targeting the proximal 5' and 3' gD untranslated sequences, essentially as described in Miyagawa et al. Wt and retargeted gD genes were then introduced by LR Clonase II-mediated recombination of the GW cassette with the different pENTR-based gD plasmids. Infectious viruses were produced by transfection of Vero cells and biological titers were determined by standard plaque assays on Vero cells. All recombinant

viruses were confirmed by DNA sequencing across the gD cassettes.

(58) Cells Murine melanoma B78H1, B78-C10 (nectin-1-transduced B78H1; Miller, C. G., et al., Development of a Syngenic Murine B16 Cell Line-Derived Melanoma Susceptible to Destruction by Neuroattenuated HSV-1, *Mol Ther* 2001, 3:160-168, doi:10.1006/mthe.2000.0240) and B78/C cells (nectin-1-transduced B78H1) (Uchida, H., et al., Generation of herpesvirus entry mediator (HVEM)-restricted herpes simplex virus type 1 mutant viruses: resistance of HVEM-expressing cells and identification of mutations that rescue nectin-1 recognition. *J Virol*, 2009. 83(7): p. 2951-61) were cultured in Dulbecco's modified Eagle's medium (DMEM; Invitrogen, Carlsbad, CA) supplemented with 5% fetal bovine serum (FBS; Invitrogen). African green monkey kidney Vero cells (ATCC CCL-81) were cultured in DMEM supplemented with 5% FBS. B78-vIII cells were established by infection of B78H1 cells with a recombinant retrovirus expressing EGFRvIII, produced by co-transfection of 293T cells (ATCC CRL-3216) with plasmids pCX4-bsr-EGFRvIII, pCL-gag-pol and pHCMV-VSVG. Infected B78H1 cultures were selected for resistance to blasticidin and resistant pools were tested for EGFRvIII expression by Western blots and flow cytometry analysis with anti-EGFR mAb H11 (ThermoFisher). After validation, the cells were sorted at the Cell Sorting Facility of the University of Pittsburgh McGowan Institute based on expression levels and clones denoted B78-vIII were isolated from a high-expressing fraction by limiting dilution. EGFRvIII expression by different B78-vIII clones is shown in FIG. 4. One clone (11 B) was used in the current study.

(59) gD Ectodomain Production and mAb Binding (SPRi) Assays. All soluble proteins used were produced in baculovirus-infected insect (Sf9) cells. All variants of HSV-1 gD(306t) were purified using a DL6 immunosorbent column as described in Sisk W P, Bradley J D, Leipold R J, et al. (High-level expression and purification of secreted forms of herpes simplex virus type 1 glycoprotein gD synthesized by baculovirus-infected insect cells. *J Virol*. 1994;68(2):766-775). Analysis of gD-mAb binding was performed using soluble proteins on the Wasatch Microfluidics CFM/SPRi system as described in Cairns T M, Ditto N T, Lou H, et al. (Global sensing of the antigenic structure of herpes simplex virus gD using high-throughput array-based SPR imaging. *PLoS Pathog*. 2017;13(6):e1006430). A CFM 2 was used to create a 48-spot microarray of amine-coupled mAbs on a CDM200M sensor chip (Xantec GmbH). Upon docking the printer chip into the SPR imager (IBIS MX96), the chip was blocked with ethanolamine and the system primed with a running buffer of PBS-0.01% Tween 20. mAb binding was assessed by flowing 100 nM soluble gD across the printed mAb array; 1M glycine pH 2.0 was used for regeneration.

(60) Fusion Assay. The fusion assay is described in Atanasiu D et al., (Dual split protein-based fusion assay reveals that mutations to herpes simplex virus (HSV) glycoprotein gB alter the kinetics of cell-cell fusion induced by HSV entry glycoproteins. *J Virol*. 2013;87(21):11332-11345) and Saw et al., (Using a split luciferase assay (SLA) to measure the kinetics of cell-cell fusion mediated by herpes simplex virus glycoproteins. *Methods*. 2015;90:68-75.) Briefly, 5×10^4 B78H1 cells (effector cells) were seeded on white, cell-culture treated 96-well plates. 4×10^5 B78-vIII cells (target cells) were seeded on 6-well plates. Transfection was performed the following day. A master mix containing 375 ng gB:NT, 125 ng each of the indicated gD construct, wt gH, wt gL, and Rluc8.sub.(1-7) was split over three wells of effector cells. Target cells were transfected with 1 μ g of Rluc8.sub.(8-11) plasmid/per well. Forty-eight hours post-transfection effector cells were pre-incubated for 1 h at 37° C. with EnduRen substrate (Promega) diluted in fusion medium (DMEM without phenol red, 50 mM HEPES and 5% FBS). Target cells were detached with versene, resuspended in fusion medium and transferred to effector cells. Fusion was triggered by the addition of target cells. A negative control (effector cells transfected with gB:NT, gH, gL, but no gD) was also included. Luciferase production was monitored over 6 h with measurements every hour using a BioTek plate reader.

(61) Blocking of fusion. Transfected effector cells were pre-incubated with both EnduRen substrate and serial, 2-fold dilutions of the indicated gD mAbs beginning at 20 μ g/ml mAbs, based on 80 μ l

final volumes.

(62) Entry Assay. Cells were infected overnight at MOI=1 and imaged for mCherry expression under a Nikon Diaphot fluorescence microscope (Nikon, Melville, NY).

(63) Real-Time PCR. Viral DNA was isolated using the Blood and Tissue DNA Extraction Kit (Qiagen, Venlo, Netherlands). Genome copy (gc) titers were then determined by creating a standard curve with an HSV-1 UL5 plasmid DNA template. The UL5 DNA sequence was amplified by PCR with primers ULSF (ACGAGCGTGGTGC GGTCATGG) (SEQ ID NO: 11) and ULSR (GCGGGTTAATAGACAATGACCACG) (SEQ ID NO: 12), and cloned into the pCR-Blunt II-TOPO vector using the Zero Blunt TOPO PCR cloning kit (ThermoFisher). A custom FAM-MGB Taqman primer probe set (ThermoFisher) was designed against the UL5 gene (UL5 qPCR F primer ATGCCGTAGTCGGCGTTTAT (SEQ ID NO: 13); UL5 qPCR R primer CGAGTTTGTCTGAGTCCATTGAC (SEQ ID NO: 14); UL5 FAM MGB probe ATGGCCAGCTCCGTAG (SEQ ID NO: 15)). Standard curves were generated for each experiment by creating a 10-fold dilution series of the UL5 plasmid (representing $3 \times 10^{6.6}$ gc corresponding to $3 \times 10^{2.2}$ gc of the HSV genome) that was amplified with an efficiency of 98-100%. Reaction conditions: 2 μ l DNA, 1 μ l of the 20 $\times$ UL5 FAM-MGB Taqman primer probe set, 10 μ l TaqMan Fast Advanced Universal PCR Master Mix (2 $\times$) in a total PCR volume of 20 μ l. Amplification conditions: 2 min at 50 $^{\circ}$ C. and 20 sec at 95 $^{\circ}$ C. for the first cycle, followed by 40 cycles of 95 $^{\circ}$ C. for 1 sec and 60 $^{\circ}$ C. for 20 sec. Samples, standard curve, and negative controls were run together in triplicate in MicroAmp Optical 96-Well Reaction Plates with the StepOne Plus Real Time PCR system (Applied Biosystems).

(64) Western Blots. Equal gc-based amounts of purified virus stocks were denatured by boiling in 1 $\times$ Laemmli buffer (Bio-Rad Laboratories, Hercules, CA) and electrophoresed on precast 4-15% SDS-PAGE gels (Bio-Rad Laboratories). Proteins were transferred to PVDF membranes (Millipore, Billerica, MA) and horizontally-cut portions of the membranes were reacted with anti-VP16 (Santa Cruz, Dallas, TX), anti-gD (DL6, Santa Cruz, Dallas, TX) or anti-gB (Virusys Corp., Taneytown, MD) prior to incubation with HRP-conjugated rabbit anti-mouse IgG (Abcam, Cambridge, UK). Membranes were developed with ECL Plus (ThermoFisher).

(65) Virus Neutralization Recombinant viruses (50-75 PFU/well) were incubated with virus-neutralizing mAbs MC5 and MC23 at a range of dilutions prior to infection of Vero cells. Infected cell monolayers were overlaid with high-density medium and plaques were counted 48 h later under the Nikon Diaphot fluorescence microscope.

(66) Results

(67) Retargeting Induces Changes in the Antigenic Structure of gD.

(68) We used SPRi analyses of purified gD ectodomains to compare the epitope landscapes of wt and retargeted gD. No binding to retargeted gD was observed for any of the yellow group mAbs (FIG. 5(A)). This was expected since these mAbs (Group VII in Table 1) recognize the HVEM binding N-terminal segment of wt gD (Table 1) that is largely replaced with our anti-EGFR/EGFRvIII scFv in the retargeted gD. In addition, we observed decreased binding to green community mAbs as well as to tan mAbs (VN members of the brown group), with some reduction among mAbs from the red and blue groups. Interestingly, all non-VN mAbs in the brown group showed increased binding (See, Cairns et al. 2017, and above, describing the various colored groups or “communities”). These observations indicated that the retargeting modifications broadly affect the conformation of gD.

(69) mar Mutations Further Change the Antigenic Structure of Retargeted gD and Reduce the Binding of Specific Neutralizing mAbs

(70) To eliminate the binding of neutralizing mAbs, we created substitution mutations P54Q and T213M in retargeted gD. Based on SPRi results, P54Q (FIG. 5(B)) completely abolished the binding of MC5 and another blue mAb, H162, that had shown increased binding to parental retargeted gD (FIG. 5(A)). This mutation also decreased the binding of two other blue mAbs as

well as of the green, brown and tan mAbs, but less dramatically (FIG. 5(B)). T213M appeared to have a more specific effect, causing mildly to severely impaired binding of the red group of mAbs, MC23 in particular, but no major changes in the binding of other mAbs (FIG. 5 (C)). Combining the two mutations in retargeted gD (P54Q/T213M) suggested an additive effect, closely resembling the binding profile of P54Q alone but with limited binding of red mAbs (FIG. 5(D)). Collectively, these results indicated that stacking of rational substitution mutations can broadly reduce the antigenic composition of retargeted gD.

(71) **mar Mutations Reduce the Virus-Free Cell-Cell Fusion Activity of Retargeted gD**

(72) We used mammalian constructs to express full-length parental and mutant retargeted gDs on the surface of HSV entry-receptor-deficient mouse melanoma B78H1 cells along with glycoproteins gB:NT and gH/gL. After incubating these cells with stably EGFRvIII-transduced B78H1 cells (B78-vIII), we measured cell-cell fusion activity by split-luciferase assay at 1-h intervals over a period of 6 h. While all retargeted constructs provoked fusion between the transfected and EGFRvIII-expressing cells, the P54Q and T213M mutants showed 50% reduced activity compared to parental gD:scEΔ38 (FIG. 6A). The P54Q/T213M double mutant was further impaired for fusion (~15% activity), indicating that the mutations either compromised presentation of the retargeted gD on the cell surface or interfered directly with the ability of the modified gDs to initiate the fusion cascade.

(73) **mar Mutations Block the Ability of Site-Specific mAbs to Inhibit Cell Fusion.**

(74) To determine whether fusion initiated by retargeted gD was sensitive to mAbs MC5 and MC23, we incubated B78H1 cells transfected with the 4 glycoprotein constructs with increasing concentrations of either mAb for 1 h before the addition of B78-vIII cells. We observed that the fusion activity of cells transfected with the parental retargeted gD construct was inhibited in a dose-dependent manner by both MC5 and MC23 (FIGS. 6B and 6C). However, the P54Q and T213M mutations reduced or completely blocked the inhibitory effects of MC5 and MC23, respectively, and the combined mutations eliminated fusion sensitivity to both mAbs (FIGS. 6B and 6C).

(75) **Virus Construction, Growth, and Entry Specificities**

(76) MC23 and MC5 are neutralizing antibodies for wt HSV-1 that appear to act at different stages of the fusion pathway, by blocking either receptor (nectin-1) binding or receptor-bound gD signaling to gH/gL, respectively. To determine whether the two mar mutations diminished retargeted virus infection, as suggested by their negative effects on cell fusion efficiency, while protecting the virus from antibodies such as MC23 and MC5, we created recombinant viruses expressing parental retargeted gD or its single or double-mutant derivatives, as described above (FIG. 3). Viruses were grown on Vero cells that naturally express both nectin-1 and simian EGFR recognized by our anti-EGFR scFv. Biological titers in PFU/μl were determined by standard plaque assays on Vero cells and physical titers in genome copies (gc)/μl were established by real-time quantitative PCR (qPCR) for the HSV-1 early gene UL5 encoding a DNA helicase-primase subunit. Comparison of gc/pfu ratios between the virus stocks suggested more efficient production of infectious virus by the unmodified retargeted virus than by its mar mutant counterparts (Table 2).

(77) **TABLE-US-00002**

TABLE 2		gD mar mutant gc number/μl		PFU/μl		gc/PFU ratios	
wt	2.30E+07						
4.20E+06	5.47	scEΔ38	1.23E+07	1.88E+06	6.54	scEΔ38-P54Q	9.49E+06
							3.04E+04
							3.12E+2
		scEΔ38-T213M	3.73E+07	7E+05	5.3E+1	scEΔ38-P54Q/T213M	5.65E+06
							1.52E+04
							3.71E+2

(78) We also tested the entry specificities of the 4 viruses by infection of cells expressing either no gD receptor (B78H1), human nectin-1 (B78/C), or EGFRvIII (B78-vIII), using Vero cells as controls. Images of mCherry expression from the common genome backbone of these viruses at 24 h post-infection showed that none of the viruses were able to enter into B78H1 or B78/C cells, while all could enter into B78-vIII and Vero cells, confirming that entry remained strictly dependent on cellular expression of primate EGFR (FIG. 7A).

(79) P54Q and P54Q/T213M mutants appeared to have lower entry efficiencies at MOI=1 compared to the parental retargeted and the T213M mutant (FIG. 7A), which may be related to the

differences in particle (gc)/PFU ratios between the stocks of the different viruses used in this particular experiment (Table 2).

(80) Comparison of gD Incorporation into Virus Particles Among Retargeted Virus Mutants.

(81) We assessed gD incorporation into purified virions by Western blot analysis of equal gc, using DL6 as the primary antibody. To control for overall differences in particle contents, we also probed the blots with antibodies to gB and the tegument protein VP16. As exemplified by the results presented in FIG. 7B, retargeting reduced gD incorporation and the mar mutations, particularly when combined, enhanced this effect. gB levels did not dramatically differ between the different viruses, including non-targeted virus (gD wt), indicating that gD and gB incorporation were mutually independent. These results demonstrated that both retargeting and the mar mutations can limit the abundance of gD in mature virions, most likely a result of impaired intracellular processing of the modified proteins, reduced stability, or both.

(82) mar Mutations Increase Resistance to Virus Neutralization by mAbs

(83) We confirmed that the retargeted virus was sensitive to inactivation by mAbs MC5 and MC23 and performed additional neutralization assays to determine whether the mar mutations offered protection against these mAbs. Viruses were incubated for 90 min with serial dilutions of MC5 or MC23 prior to infection of Vero cells, and plaques were counted at 48 h. Panels A and B of FIG. 8 show that each mar mutation increased the resistance of the retargeted virus in a manner consistent with the earlier binding and fusion-inhibition results. Specifically, the P54Q mutant showed resistance to mAb MC5, and similarly, the T213M mutant was resistant to MC23. Furthermore, the P54Q/T213M double mutant was resistant to both mAbs (FIG. 8 (C, D)).

(84) We expanded these results by examining the resistance of the double-mutant retargeted virus to VN by two other mAbs, H162 from the same group as MC5 (blue), and LP2 from the same group as MC23 (red). H162, whose binding to retargeted gD was abolished by the double mutation (FIG. 5(D)), neutralized the retargeted virus while the double mutant was fully protected (FIG. 9(A)). Similarly, the mutations caused increased resistance to LP2 although some remaining sensitivity at the lower mAb dilutions was evident (FIG. 9(B)), consistent with low residual binding of LP2 to the double-mutant retargeted gD (FIG. 5(D)). Thus, the effects of the mutations on virus neutralization were not limited to single representatives of the blue and red mAb groups.

(85) Since the SPRi results revealed increased binding of all or nearly all of the mAbs of the brown group to retargeted and double-mutant retargeted gD compared to wt gD (FIG. 5 (A, D)), we tested 2 of the brown mAbs for neutralization of viruses containing these different versions of gD. The results (FIG. 9 (C, D)) showed that both mAbs were neutralizing for both of the retargeted viruses, while the non-retargeted virus expressing wt gD was substantially less sensitive to either mAb. We suggest that these mAbs gain neutralizing activity through increased accessibility of their epitope(s) caused by the retargeting modifications and that the P54Q/T213M double mutation does not restore the protected wt conformation of this region to reduce or reverse this effect.

(86) It should be noted that human immune sera typically do not block the binding of mAbs from the brown group (e.g., MC14) to wt gD (Cairns et al, 2015), suggesting that HSV-immune individuals do not contain antibodies to the brown epitope(s) of retargeted gD.

(87) The present invention has been described with reference to certain exemplary embodiments, dispersible compositions and uses thereof. However, it will be recognized by those of ordinary skill in the art that various substitutions, modifications or combinations of any of the exemplary embodiments may be made without departing from the spirit and scope of the invention. Thus, the invention is not limited by the description of the exemplary embodiments, but rather by the appended claims as originally filed.

Claims

1. A retargeted herpes simplex virus (HSV) particle, comprising a genome, a capsid, tegument, and an envelope comprising an antigenically-modified glycoprotein D (gD) protein, wherein one or more of amino acids 10-20, 54, 75-79, 132, 140, 213, 216, 222-224, and 262-279 of SEQ ID NO: 1 or SEQ ID NO: 2, or one or more amino acids corresponding to amino acids 10-20, 54, 75-79, 132, 140, 213, 216, 222-224, and 262-279 of SEQ ID NO: 1 or SEQ ID NO: 2, are modified in the gD glycoprotein of the virus particle, and wherein the modifications: reduce or eliminate neutralization of the retargeted virus particle by antibodies that recognize the gD glycoprotein, and permit gD glycoprotein-mediated infection of a target cell by the retargeted virus particle.
2. The virus particle of claim 1, wherein one or both of amino acids P54 and T213 of SEQ ID NO: 1 or SEQ ID NO: 2, or one or both amino acids corresponding to amino acids P54 and T213 of SEQ ID NO: 1 or SEQ ID NO: 2 are modified.
3. The virus particle of claim 1, further comprising an antigenically-modified glycoprotein B (gB), wherein the modification is in one or more of amino acids 47, 62, 85, 203, 303, 304, 305, 308, 328, 335, 419, 473, 594, or 640-670 of SEQ ID NO: 3 or amino acid 412 of SEQ ID NO: 4, or one or more amino acids corresponding to amino acids 47, 62, 85, 203, 303, 304, 305, 308, 328, 335, 419, 473, 594, or 640-670 of SEQ ID NO: 3 or amino acid 412 of SEQ ID NO: 4.
4. The virus particle of claim 1, having a 438 mutation of SEQ ID NO: 1 or 2, or a mutation in a gD glycoprotein corresponding to a 438 mutation of SEQ ID NO: 1 or 2.
5. The virus particle of claim 1, comprising a non-native ligand capable of binding a surface component of a target cell type.
6. The virus particle of claim 5, wherein the target cell type is a cancer cell.
7. The virus particle of claim 5, wherein the non-native ligand capable of binding a surface component of a target cell type is incorporated into a viral envelope glycoprotein of the virus particle.
8. The virus particle of claim 7, wherein the viral envelope glycoprotein of the virus particle into which the ligand is incorporated is gD, gC, gB and/or gH.
9. The virus particle of claim 5, wherein the surface component is one or more of EGFR, EGFRvIII, other oncogenic EGFR variants, HER2, CD133, CXCR4, carcinoembryonic antigen (CEA), CLC-3/annexin-2/MMP-2, human transferrin receptor, EpCAM, or c-Met.
10. The virus particle of claim 1, wherein binding of at least one or more viral envelope glycoproteins to its natural receptor is eliminated.
11. The virus particle of claim 1, wherein the genome comprises an exogenous expression cassette.
12. The virus particle of claim 11, wherein the exogenous expression cassette encodes an agent that enhances tumor killing activity.
13. The virus particle of claim 1, wherein the genome comprises a target sequence for one or more microRNAs.
14. A recombinant HSV genome encoding the virus particle of claim 1.
15. A viral stock, comprising at least 10⁵ pfus of a virus particle of claim 1.
16. A pharmaceutical dosage form comprising at least 10⁵ pfus of a virus particle of claim 1, and a pharmaceutically-acceptable excipient.
17. The dosage form of claim 16, formulated for parenteral delivery.
18. A method of treating a patient having a cancer, comprising administering to the patient an amount of the virus particle of claim 6 effective to treat a cancer patient.
19. The virus particle of claim 2, wherein the modifications to SEQ ID NO: 1 or SEQ ID NO: 2 comprise P54Q and/or T213M.
20. A retargeted herpes simplex virus (HSV) particle, comprising a genome, a capsid, tegument, and an envelope comprising an antigenically-modified glycoprotein D (gD) protein, wherein the antigenically-modified gD protein comprises a P54Q and/or a T213M substitution in SEQ ID NO: 1 or SEQ ID NO: 2 and optionally one or more of amino acids 10-20, 75-79, 132, 140, 216, 222-

224, and 262-279 of SEQ ID NO: 1 or SEQ ID NO: 2, or one or more amino acids corresponding to amino acids 10-20, 75-79, 132, 140, 216, 222-224, and 262-279 of SEQ ID NO: 1 or SEQ ID NO: 2, are modified in the gD glycoprotein of the virus particle, and wherein the modifications: reduce or eliminate neutralization of the retargeted virus particle by antibodies that recognize the gD glycoprotein, and permit infection of a target cell by the retargeted virus particle.

21. The virus particle of claim 20, further comprising an antigenically-modified glycoprotein B (gB), wherein the modification is in one of more of amino acids 47, 62, 85, 203, 303, 304, 305, 308, 328, 335, 419, 473, 594, or 640-670 of SEQ ID NO: 3, or amino acid 412 of SEQ ID NO: 4, or one or more amino acids corresponding to amino acids 47, 62, 85, 203, 303, 304, 305, 308, 328, 335, 419, 473, 594, or 640-670 of SEQ ID NO: 3, or amino acid 412 of SEQ ID NO: 4.
